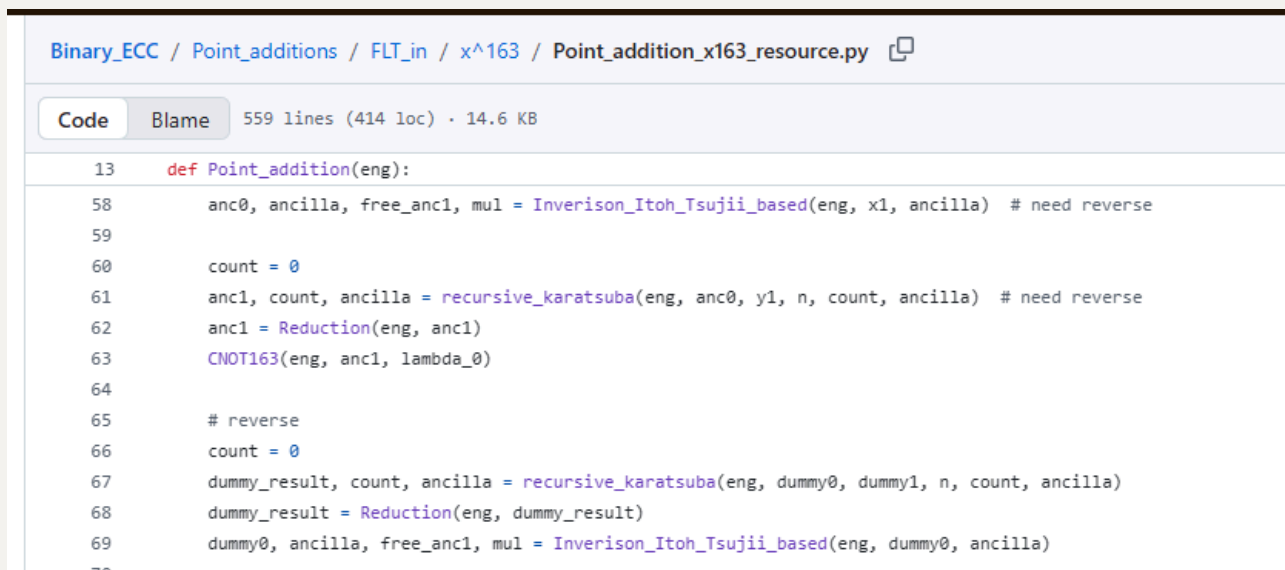


We acknowledge that the previous wording confused two distinct issues. The specific ProjectQ problem discussed in [HZL25] does not itself affect the circuit width.

However, upon re-examining the public Binary_ECC implementation accompanying [JSB+25], we discovered another issue in the circuit model used by the FLT-in resource estimation script. Some uncompute operations marked as inversions are represented by calling `uncompute` on individual virtual registers instead of performing corresponding `uncompute` operations on the original data register and working register. This construction can serve as an alternative to gate count statistics, but it fails to preserve the qubit-correlated DAG of the intended circuit.



```
Binary_ECC / Point_additions / FLT_in / x^163 / Point_addition_x163_resource.py
Code Blame 559 lines (414 loc) · 14.6 KB

13 def Point_addition(eng):
58     anc0, ancilla, free_anc1, mul = Inverison_Itoh_Tsujii_based(eng, x1, ancilla) # need reverse
59
60     count = 0
61     anc1, count, ancilla = recursive_karatsuba(eng, anc0, y1, n, count, ancilla) # need reverse
62     anc1 = Reduction(eng, anc1)
63     CNOT163(eng, anc1, lambda_0)
64
65     # reverse
66     count = 0
67     dummy_result, count, ancilla = recursive_karatsuba(eng, dummy0, dummy1, n, count, ancilla)
68     dummy_result = Reduction(eng, dummy_result)
69     dummy0, ancilla, free_anc1, mul = Inverison_Itoh_Tsujii_based(eng, dummy0, ancilla)
70
```

Therefore, the resulting depth is not an efficient estimate of the intended circuit and may be underestimated. This is a circuit modeling problem in the Binary_ECC resource script, not a defect in ProjectQ itself.

Furthermore, our code forcibly places toffoli of the same depth sequentially on the gates to calculate the toffoli depth, resulting in different depths, but this is not an error.